Exp No:7
2026

Frequency Response of Series and Parallel Resonance circuit

AIM:

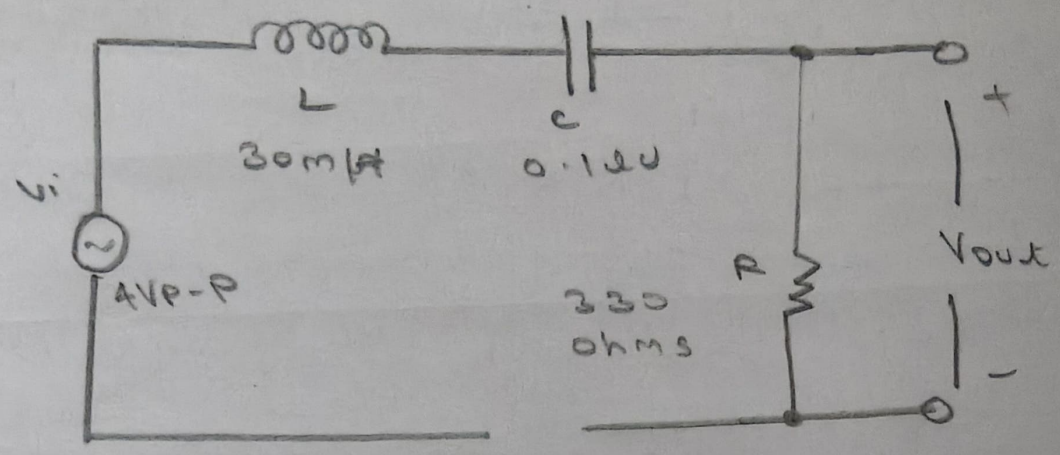
To Study the behaviour of Series and Parallel RLC circuit at resonance and to determine the resonant frequency bandwidth and Q of the given RLC circuit using multisim Simulator

Apparatus Required

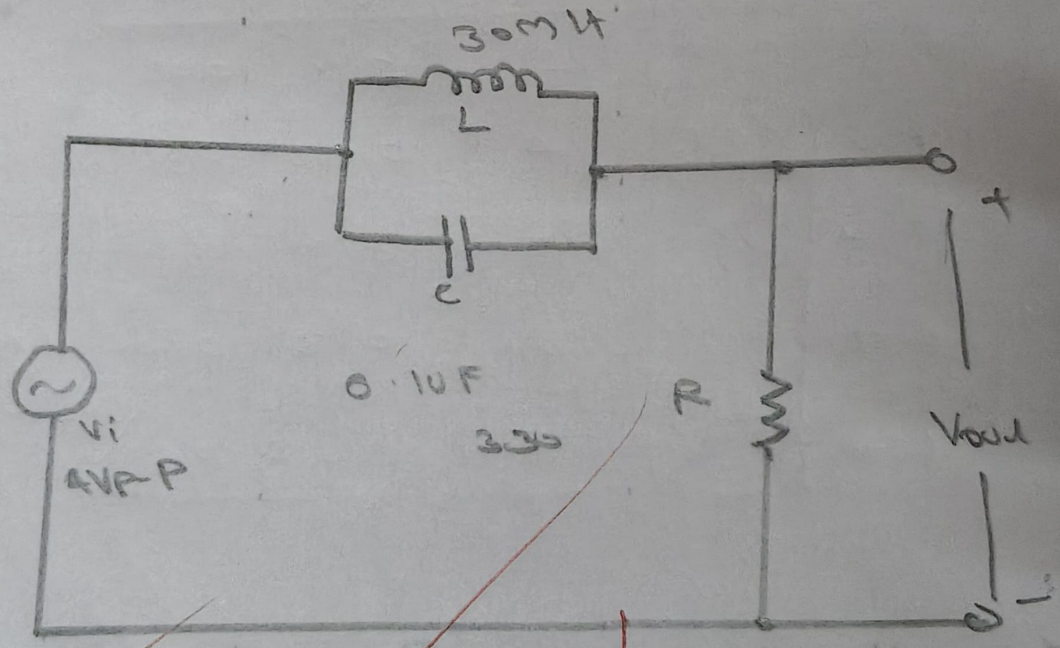
S.No	Components	Specification	Quantity
1.)	CRO	10MHz	1
2.)	DIB	–	1
3.)	capacitors	0.1 μF	1
4.)	Resistors	330 Ω	1
5.)	BreadBoard	–	–
6.)	Connecting wires	–	–
7.)	PC with multisim Simulator	–	–

CIRCUIT DIAGRAM :

(i)



(ii)



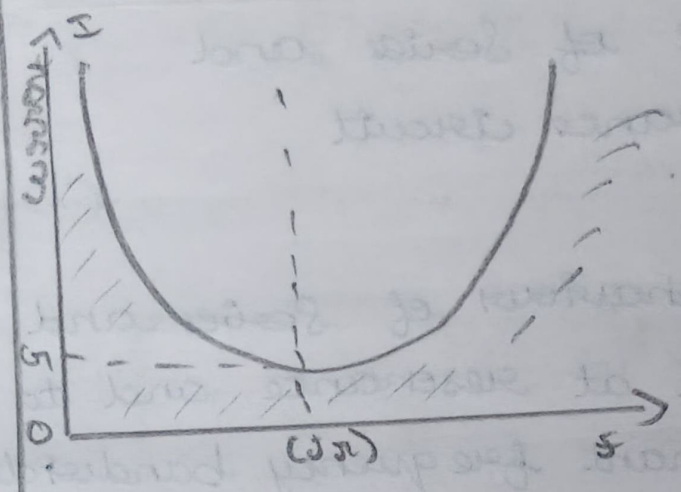
Formular :

$f_0 = 1/2\pi\sqrt{LC}$

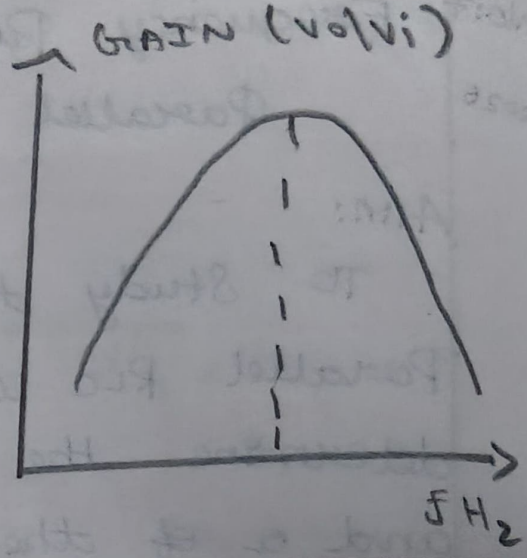
$BW = f_2 - f_1$

$Q = f_0/BW$

PARRLLEL RESONANCE :.



SERIES RESONANCE :.



TABULATION :.

SERIES RESONANCE			PARALLEL RESONANCE		
frequency (Hz)	output voltage (v)	output current (mA)	frequency (Hz)	output voltage (v)	output current (mA)
500	1.2	2	500	1.5	9
700	2.0	4	700	2.8	7
900	3.5	7	900	4.0	5
1000	5.0	10	1000	5.0	2
1100	3.6	7	1100	4.1	5

Verified

Procedure

1) Construct circuit shown in Figure 1on bread board

2) connect CRO Ch1 to input and ch2 to output

3) Set the input voltage to 4VP-P

4) vary the frequency from 100Hz to 3KHz in small steps to get a maximum output voltage magnitude. The Frequency at this maximum voltage Vm is the resonance frequency

5) calculate the bandwidth by subtracting the 2-cut off frequencies.

6) Calculate the Q factor

7) Repeat ~~to~~ Step 1 through 7 for circuit show in figure 2

Marks SPLIT UP

S.NO	Description	Marks allotted	Marks obtained
1.)	Experimental / Setup / connections	20 marks	20
2.)	observation/readings	20 marks	20
3.)	calculation/Result	20 marks	20
4.)	Viva	20 marks	20
5.)	Result	20 marks	20
			100

Calculation:

$L = 100mH = 0.1 H$

$C = 0.25 \mu F = 0.25 \times 10^{-6} F$

$$F_0 = \frac{1}{2\pi\sqrt{(0.1)(0.25 \times 10^{-6})}}$$

$$F_0 = \frac{1}{2\pi\sqrt{1.58 \times 10^{-4}}}$$

$$F_0 = \frac{1}{9.92 \times 10^{-4}}$$

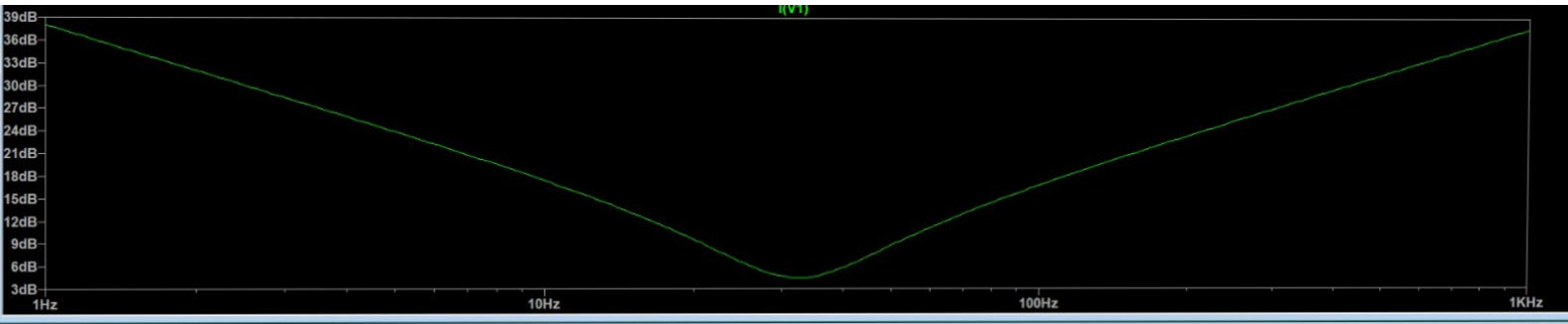
$F_0 = 1000 Hz$

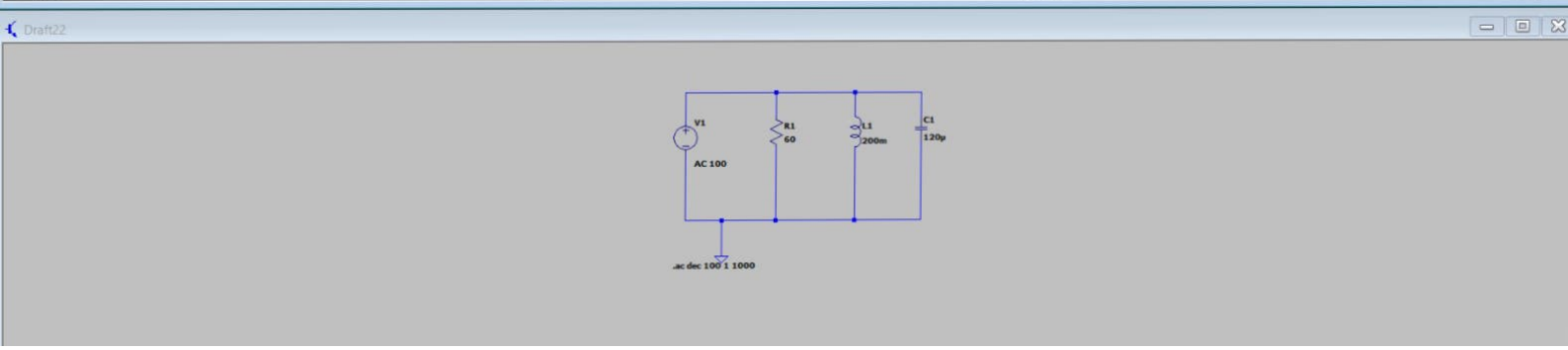
2) maximum current (series) and minimum current (parallel) occur at 1000 Hz.

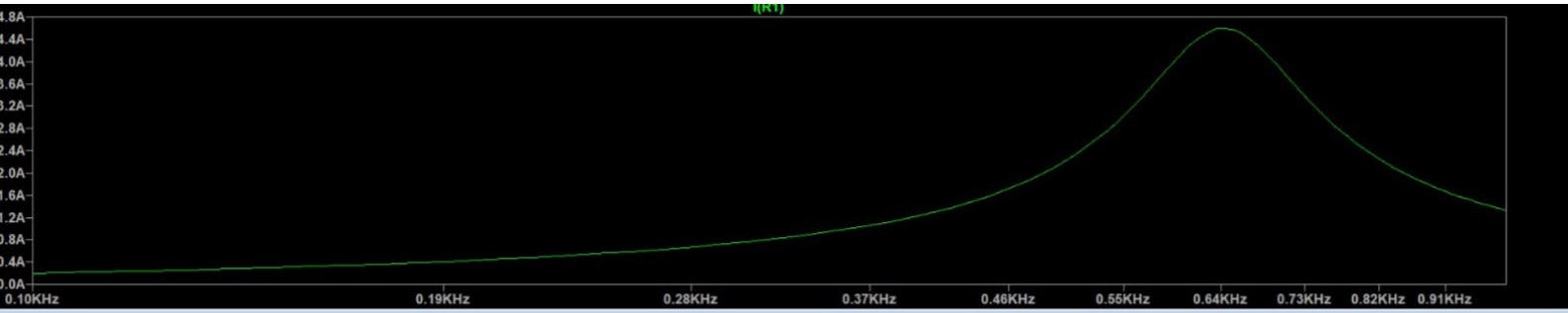
RESULT :-

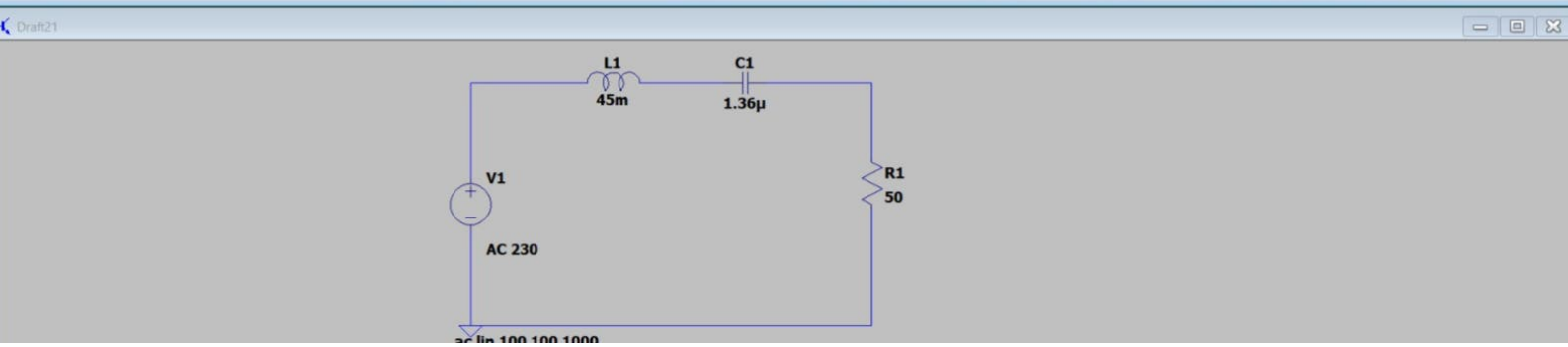
Thus the phenomenon of resonance in RLC circuit was studied and the following were determined using multisim simulator.

Series RLC circuit :-	Parallel RLC circuit :-
Resonant frequency :-	Resonant frequency =
Bandwidth =	Bandwidth =
Q factor =	Q factor =


I(V1)
39dB
36dB
33dB
30dB
27dB
24dB
21dB
18dB
15dB
12dB
9dB
6dB
3dB
1Hz
10Hz
100Hz
1KHz


Draft22
V1
AC 100
R1
60
L1
200m
C1
120µ
.ac dec 100 1 1000


I(R1)
4.8A
4.4A
4.0A
3.6A
3.2A
2.8A
2.4A
2.0A
1.6A
1.2A
0.8A
0.4A
0.0A
0.10KHz
0.19KHz
0.28KHz
0.37KHz
0.46KHz
0.55KHz
0.64KHz
0.73KHz
0.82KHz
0.91KHz


Draft21
L1
45m
C1
1.36µ
V1
AC 230
R1
50